

CLOUD MLOPS & AI PLATFORM

Azure Fundamentals & Azure ML Workspace

Navigating Azure Cloud Services, Architecting Workspaces, & Scaling Enterprise MLOps

Day 2 | Session 09 (9:00 AM – 10:00 AM) • Azure ML SDK v2 • Auto-Scaling Compute & MLflow

Why Move Machine Learning to Azure?

Overcoming local workstation limitations with enterprise cloud infrastructure

Local Workstation Bottlenecks

- **Limited Hardware & VRAM**

Constrained by laptop CPU/GPU RAM. Unable to train multi-gigabyte models.

- **Zero Auto-Scaling**

Training jobs block local machines for hours with zero elasticity.

- **Siloed Artifacts & Data**

Metrics and models live in local folders without governance or versioning.

- **Production Friction**

Rewriting local scripts for production servers leads to silent bugs.

The Azure ML Cloud Solution

- **Elastic Compute Clusters**

Provision multi-GPU clusters in minutes that auto-scale to zero when idle.

- **Single Pane of Glass**

Unified workspace centralizing datasets, jobs, metrics, and model registry.

- **Automated Lineage & MLflow**

Native MLflow tracking logs every parameter, artifact, and code version.

- **Managed Real-Time Endpoints**

One-click deployment to real-time REST APIs with safe blue/green rollout.

Navigating Azure Services: Core Hierarchy

How Azure organizes billing, security boundaries, and ML resources

1. Azure Subscription — Master Billing & Resource Quotas

Defines the financial account boundary and enterprise quotas for GPU/CPU cores.

2. Resource Group — Project & Lifecycle Namespace

Logical folder grouping all related resources (AML Workspace, Storage, Key Vault, ACR).

3. Region — Physical Geographic Data Center

Geographic data center location (e.g., East US, Southeast Asia) selected for latency & compliance.

4. Azure ML Workspace — Central Hub for Machine Learning

The top-level resource organizing all machine learning assets, compute, and deployments.

The Core Infrastructure Quartet

Four essential Azure companion services automatically provisioned with your Workspace

Azure Storage Account

Role: Blob Storage & Data Lake

Default datastore storing raw datasets, model weights (model.pkl), and job output artifacts.

Azure Key Vault

Role: Secrets & Credentials Locker

Securely manages database connection strings, S3 secrets, and API access tokens with zero hardcoding.

Container Registry (ACR)

Role: Docker Image Repository

Stores customized Docker images used to execute distributed training and containerized inference.

Application Insights

Role: Live Telemetry Watchtower

Monitors deployed online endpoints in real time, capturing request rates, latency, and error traces.

Azure ML Workspace: Asset Anatomy

A unified control plane organized into three functional operational pillars

Authoring Plane

- **Notebooks**

Interactive cloud Jupyter development

- **Automated ML**

No-code/low-code algorithm search

- **Designer**

Visual drag-and-drop pipeline graph builder

Asset Management Plane

- **Data Assets**

Versioned datasets with lineage

- **Jobs & Pipelines**

MLflow-tracked execution DAGs

- **Environments**

Reproducible Conda + Docker specs

- **Model Registry**

Versioned models with stage gates

Management & Serving

- **Compute Targets**

Instances, clusters, and serverless

- **Online Endpoints**

Low-latency REST APIs (Blue/Green)

- **Batch Endpoints**

Asynchronous bulk scoring engines

- **RBAC & Security**

Enterprise access control & VNet

Azure ML Compute Engines & Cost Strategies

Selecting the right compute target to maximize throughput and eliminate idle costs

Compute Instance

Scaling:

Single Dedicated VM (1 node)

Billing Model:

Billed 100% while Running

Best Suited For:

Interactive EDA & debugging

Compute Cluster

Scaling:

Auto-scales 0 □ N nodes

Billing Model:

ONLY during active training

Best Suited For:

Production training & sweeps

Serverless Compute

Scaling:

Managed Dynamic Execution

Billing Model:

Billed per second of job

Best Suited For:

Ad-hoc jobs with zero setup

Managed Endpoint

Scaling:

Auto-scales on Traffic (QPS)

Billing Model:

Billed per active replica VM

Best Suited For:

Real-time low-latency REST API

Cost Optimization Golden Rule: Auto-Scale to Zero (min_instances = 0)

Set min_instances = 0 and idle_time_before_scale_down = 120s on training clusters. Pay \$0.00 when jobs finish!

Datastores vs. Data Assets

Connecting to cloud data sources and creating versioned, reproducible dataset snapshots

Datastore (Storage Connection)

Defines secure authentication connection to storage services (Blob, ADLS Gen2, SQL).

Encapsulates credentials (SAS token, Service Principal) without exposing them in code.

Default workspace datastore: workspaceblobstore (automatically mounted).

Data Asset (Versioned Snapshot)

A versioned, metadata-rich pointer to specific files/folders within a datastore.

Guarantees exact training reproducibility: Model trained on Dataset v2 can always be audited.

Supports tabular (pandas/parquet) and file/folder asset formats with lineage tracking.

Reproducible Environments with Azure ML

Decoupling execution environments from compute hardware using Docker + Conda

1. Curated Environments — Pre-built by Microsoft

Pre-packaged Docker images containing PyTorch, TensorFlow, Scikit-learn, and CUDA drivers. Zero build time.

2. Custom Conda Environments — Conda YAML over Base OS

Specify your custom Python libraries (requirements.txt / conda.yaml) layered over a lightweight Ubuntu base image.

3. Custom Docker Images — Full OS & C++ Control

Provide a custom Dockerfile with proprietary system libraries. Azure builds and caches the image in Azure Container Registry (ACR).

Model Registry & MLflow Integration

Tracking experiments and transitioning models across governance lifecycle stages

1. Experiment Run

MLflow autologging tracks parameters, loss curves, confusion matrices, and artifacts.

2. Registration

Top performing model is registered with version tag (e.g., fama_risk_model:v3).

3. Staging Gate

Automated validation tests data drift, fairness metrics, and latency constraints.

4. Production Serving

Model is promoted to Production stage and deployed to Managed Online Endpoint.

Inference Endpoints: Online vs. Batch

Exposing models for real-time predictions or scalable asynchronous bulk processing

Managed Online Endpoints

- **Use Case:**

Real-time, low-latency REST APIs for web apps, mobile clients, and microservices.

- **Deployment:**

Fully managed infrastructure with auto-scaling based on CPU or Request queue.

- **Traffic Splitting:**

Safe Blue/Green deployments (e.g. 90% traffic to v1, 10% canary to v2).

- **Security:**

Secured via Key authentication, Azure AD tokens, and VNet private endpoints.

Batch Endpoints

- **Use Case:**

Asynchronous, high-throughput batch scoring over millions of historical records.

- **Compute Elasticity:**

Automatically spins up compute nodes upon trigger, executes, and scales to 0.

- **Cost Efficiency:**

Zero idle cost. Pay only for the exact compute time taken to score the dataset.

- **Output Routing:**

Results automatically written to Azure Blob Storage, SQL, or Lakehouse.

Connecting via Azure ML Python SDK v2

Clean, object-oriented resource management and authentication

```
from azure.ai.ml import MLClient
from azure.identity import DefaultAzureCredential
from azure.ai.ml.entities import AmlCompute

# 1. Authenticate (Env Vars -> Managed Identity -> Azure CLI -> Browser)
credential = DefaultAzureCredential()

# 2. Connect to Azure ML Workspace
ml_client = MLClient(
    credential=credential,
    subscription_id='sub-fama-prod-001',
    resource_group_name='rg-fama-mlops-eastus',
    workspace_name='aml-fama-workspace'
)
print(f'Connected to: {ml_client.workspace_name}')

# 3. Define Auto-Scaling Training Compute (0 -> 4 nodes)
cpu_cluster = AmlCompute(
    name='cpu-cluster-train',
    type='amlcompute',
    size='Standard_DS12_v2',
    min_instances=0,    # Scale to 0 when idle = $0.00 cost
    max_instances=4,
    idle_time_before_scale_down=120
)
ml_client.compute.begin_create_or_update(cpu_cluster)
```

On-Premise vs. Azure Machine Learning

Direct operational and financial capability breakdown

Operational Capability	On-Premise Infrastructure	Azure ML Enterprise Platform
Compute Provisioning	Weeks to procure & configure physical servers	Minutes to spin up multi-node GPU clusters
Scaling & Utilization	Fixed capacity; servers sit idle overnight	Elastic auto-scale to zero (min_instances = 0)
Experiment Tracking	Manual spreadsheets or self-hosted servers	Native MLflow autologging & model lineage
Deployment & Serving	Complex manual Docker & NGINX setups	Managed Online Endpoints with Blue/Green routing
Cost Model	CAPEX (large upfront hardware expenditure)	OPEX (pay-per-second only while jobs run)

Key Takeaways: Navigating Azure for MLOps

Core principles for building scalable, cost-effective cloud AI systems

1. Workspace is the Single Pane of Glass

Centralizes compute, datasets, experiment tracking, and model versions for complete traceability.

2. Scale-to-Zero is Mandatory for Cost Governance

Always set min_instances = 0 on training clusters so compute charges stop the moment jobs complete.

3. Leverage Companion Services

Blob Storage holds data, Key Vault secures credentials, ACR stores containers, and App Insights monitors traffic.

4. Standardize on SDK v2 & Declarative YAML

Use modern SDK v2 entities compatible with MLOps CI/CD pipelines (GitHub Actions / Azure DevOps).